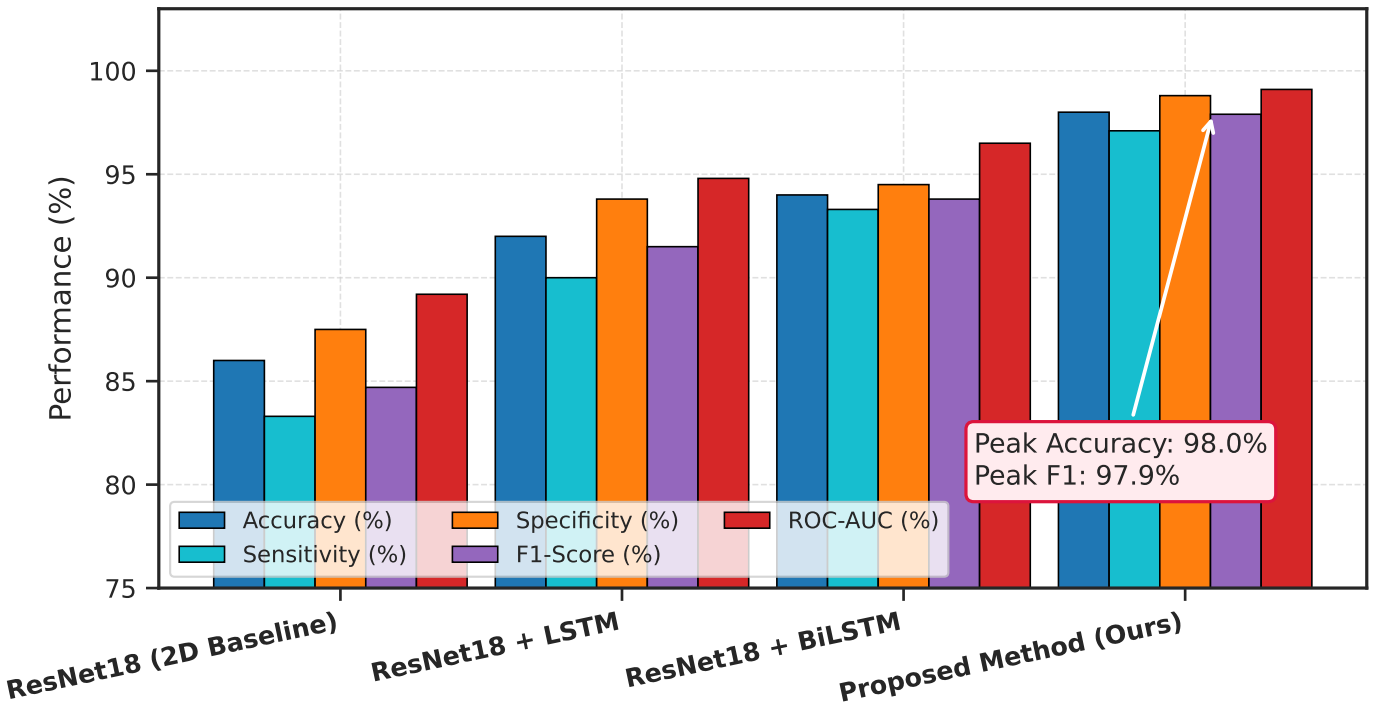


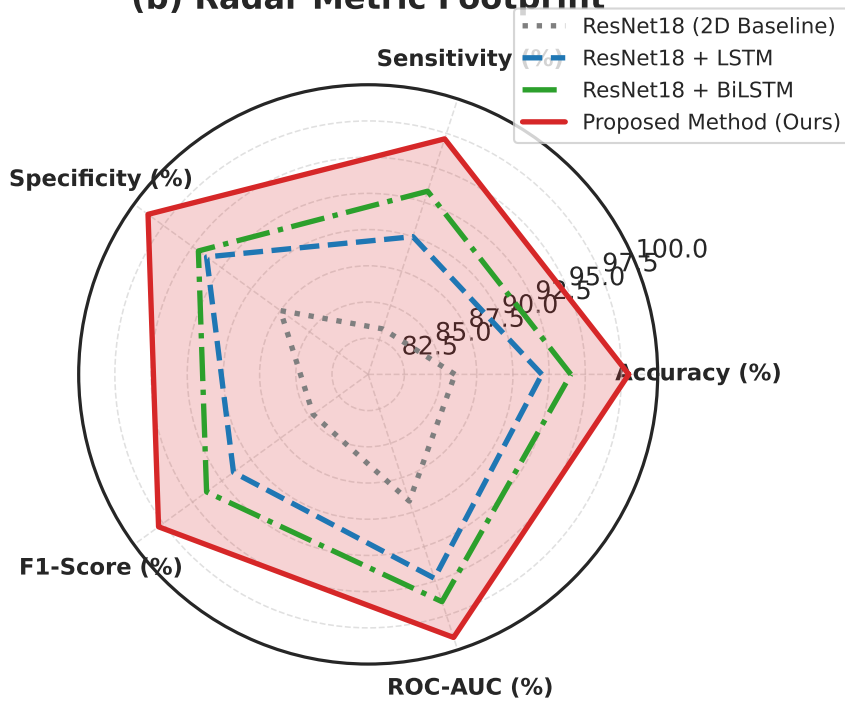


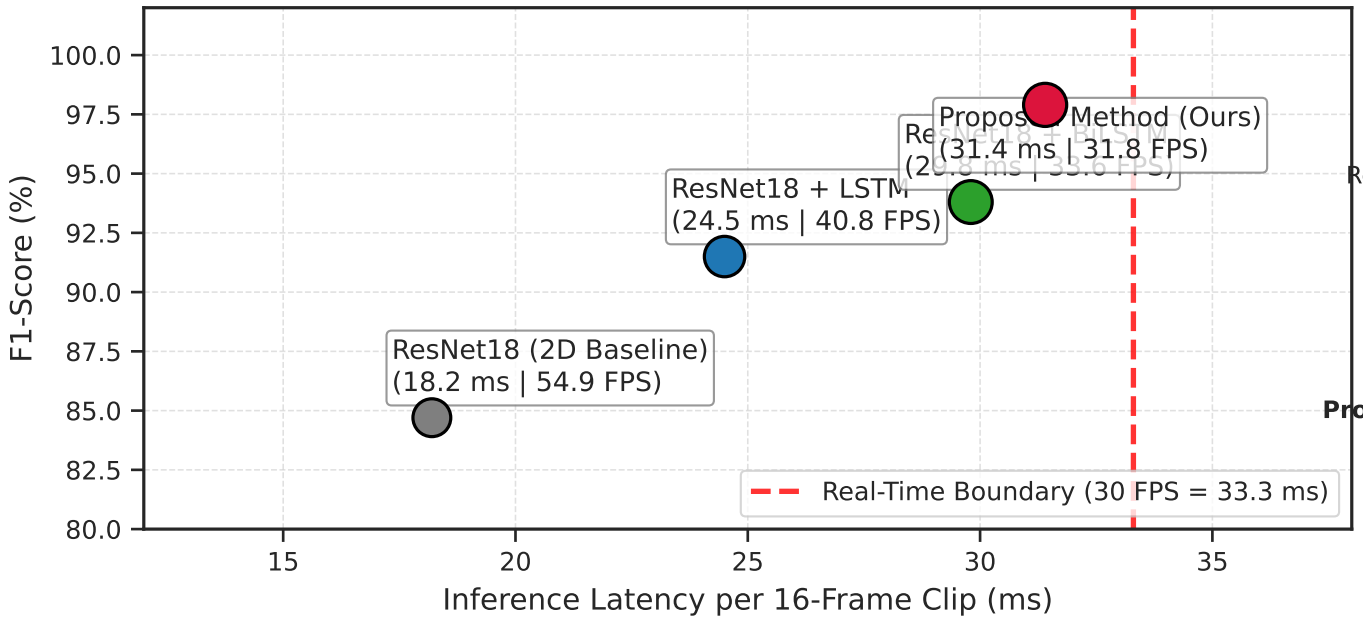


## (d) Computational Complexity & Deployment Feasibility

| Architecture | Params | Latency | FPS | F1 |
|---|---|---|---|---|
| ResNet18 (2D Base | 11.18M | 18.2 ms | 54.9 | 84.7% |
| ResNet18 + LST | 12.76M | 24.5 ms | 40.8 | 91.5% |
| ResNet18 + BiLS | 14.33M | 29.8 ms | 33.6 | 93.8% |
| **Proposed Method (** | **14.53M** | **31.4 ms** | **31.8** | **97.9%** |